

12.421 / 12.621 Lecture 4

EM wave transmission and reflection at
planar interfaces: Part 1,

Normal Incidence

Topics covered

- Transverse waves on strings as an analogy for EM waves
- Boundary Conditions for transverse waves
- EM waves w/ normal incidence to a plane
- EM boundary conditions
- Fresnel Eqs for normal incidence

- Transverse waves on a string

- Let's start w/ some intuition building

- Consider canonical example of waves on strings

2 types

- longitudinal waves (e.g., sound waves, stretching of Slinky, etc.)

- transverse waves (e.g., waves generated by shaking or plucking strings)

↳ EM waves are always transverse

- Recall wave eq for wave traveling in z direction

$$\frac{\partial^2 f}{\partial z^2} = \frac{1}{v_p^2} \frac{\partial^2 f}{\partial t^2}$$

- For a string, you may recall from physics classes that

$$v_p = \sqrt{\frac{T}{m}} \quad \text{where } T \equiv \text{tension of string}$$

$m \equiv \text{mass per unit length}$

(for EM waves, we showed before that $v_p = \frac{c}{n}$)
 where $n' = kc\sqrt{\epsilon}$

- Let us consider two strings attached at $z=0$;
 string 1 extends to $z \rightarrow -\infty$ and string 2 to $z \rightarrow \infty$

- $T_1 = T_2$

- $m_1 \neq m_2 \Rightarrow v_{p1} \neq v_{p2}$ (just as we'd expect for EM waves when $\epsilon_1 \neq \epsilon_2$)

• Trans waves on string (cont.)

- incident wave in $+z$ direction

$$f_I = A_I e^{j(k_1 z - \omega t)} \quad (\text{string 1, } z < 0)$$

- gives a reflected wave (traveling in $-z$ direction in string 1)

$$f_R = A_R e^{j(-k_1 z - \omega t)} \quad (z < 0)$$

- and a transmitted wave (traveling in $+z$ direction in string 2)

$$f_T = A_T e^{j(k_2 z - \omega t)} \quad (\text{string 2, } z > 0)$$

- Now, the frequency ω must be the same everywhere and is set by the source (e.g., person shaking the string, RS instrument, etc.)

- Recalling the dispersion relation

$$\omega = K v_p \rightarrow \text{if } m_1 \neq m_2, \text{ then } v_{p1} \neq v_{p2} \text{ and } k_1 \neq k_2$$

$\Rightarrow$ wavelengths must differ if $m_1 \neq m_2$

- Now, let us define the net displacement in each string

$$(*) \quad f(z, t) = \begin{cases} A_I e^{j(k_1 z - \omega t)} + A_R e^{j(-k_1 z - \omega t)} & \text{for } z < 0 \\ A_T e^{j(k_2 z - \omega t)} & \text{for } z > 0 \end{cases}$$

• Trans waves on strings (cont.)

- Define BCS

$$\text{At } z=0$$

1. $f(z=0^-, t) = f(z=0^+, t)$ otherwise, strings would be disconnected

2. $\left. \frac{\partial f}{\partial z} \right|_{0^-} = \left. \frac{\partial f}{\partial z} \right|_{0^+}$ otherwise there would need to be a net force acting at $z=0$

- Applying BCS to Eq. ~~8~~, we have

$$A_I + A_R = A_T \quad \text{and} \quad K_1(A_I - A_R) = K_2 A_T$$

- In RS, we often know A_I , so rearrange to get

$$A_R = \left(\frac{K_1 - K_2}{K_1 + K_2} \right) A_I = \left(\frac{v_{P2} - v_{P1}}{v_{P2} + v_{P1}} \right) A_I$$

and

$$A_T = \left(\frac{2K_1}{K_1 + K_2} \right) A_I = \left(\frac{2v_{P2}}{v_{P2} + v_{P1}} \right) A_I$$

- As we went over w/ EM waves, A_I, A_R, A_T can be complex-valued, and we can write them in terms of magnitude and phase ($\delta_I, \delta_R, \delta_T$) as

$$|A_R| e^{j\delta_R} = \left(\frac{v_{P2} - v_{P1}}{v_{P2} + v_{P1}} \right) |A_I| e^{j\delta_I} \quad (**)$$

$$|A_T| e^{j\delta_T} = \left(\frac{2v_{P2}}{v_{P2} + v_{P1}} \right) |A_I| e^{j\delta_I}$$

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• Trans waves on strings

- Important takeaways from Egs **

• Continuity requires $\sigma_T = \sigma_I$

• But ** requires

$$\sigma_R = \begin{cases} \sigma_I - \pi & \text{for } v_{p1} > v_{p2} \text{ (reflected wave upside down)} \\ \sigma_I & \text{for } v_{p2} > v_{p1} \text{ (reflected wave is right side up)} \end{cases}$$

• If $m_2 \gg m_1$ (sim. if string 2 is fixed in place or, in the EM case $\sigma_2 \gg \sigma_1$)

$$v_{p2} \ll v_{p1} \Rightarrow |A_R| \approx |A_I| \text{ and } |A_T| \approx 0$$

• EM Boundary Conditions

- Having developed our intuition using the string analog,

let's move to EM waves

- First, let's derive the BCs

• Maxwell's Egs in integral form

$$\begin{aligned} \oint_S \epsilon' \epsilon_0 \underline{E} \cdot d\underline{a} &= 0 & \text{Gauss's Law} \\ \oint_S \mu \underline{H} \cdot d\underline{a} &= 0 & \text{Gauss's Law for mag} \end{aligned}$$

line integral over closed loop

any closed surface S with normal $\underline{a}$ (positive pointing from medium 2 to medium 1)

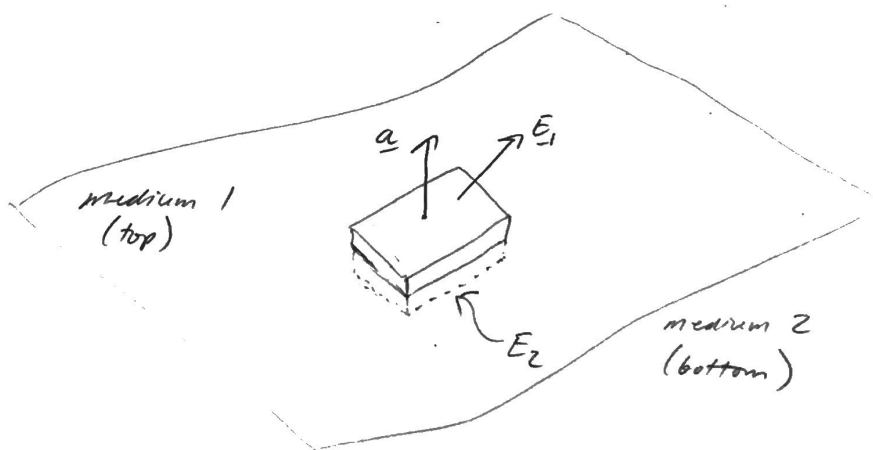
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• EM Waves (cont.)

$$\left. \begin{aligned} \oint_P \underline{E} \cdot d\underline{\ell} &= -\mu \int_S \frac{d\underline{H}}{dt} \cdot d\underline{a} & \text{Faraday's Law} \\ \oint_P \underline{H} \cdot d\underline{\ell} &= I_f + \epsilon \epsilon_0 \int_S \frac{d\underline{E}}{dt} \cdot d\underline{a} & \text{Ampere's Law} \end{aligned} \right\} \begin{array}{l} \text{any surface } S \\ \text{bounded by closed} \\ \text{loop } P \end{array}$$

where $I_f \equiv$ free current passing through loop, taken to be $I_f = 0$ hereafter

- Gauss's Law: - Let's consider a very thin pill box centered on the interface between two media and extending slightly into each



we have from Gauss's Law

$$\epsilon_1' \underline{E}_1 \cdot \underline{a} = \epsilon_2' \underline{E}_2 \cdot \underline{a}$$

(note that the edges of our pillbox contribute nothing as the thickness of the box $\rightarrow 0$)

Since $\underline{a}$ is the normal

$$\boxed{\epsilon_1' E_1^\perp = \epsilon_2' E_2^\perp}$$

(where $E_i^\perp$ denotes the component of $\underline{E}_i$ field perpendicular to the interface)

• EM waves (cont)

- BCs (cont)

- Gauss's Law for Magnetism: following similar logic as for Gauss's Law for electricity, we have

$$\mu_1 H_1^\perp = \mu_2 H_2^\perp$$

Given that in RS we nearly always have

$\mu = \mu_0$, can simplify to

$$\boxed{H_1^\perp = H_2^\perp}$$

$\Rightarrow$ Normal component of magnetic field is conserved

- Faraday's Law: Now imagine our pill box is very narrow in one horizontal, and the long side has length l , we have

$$(\underline{E}_1 - \underline{E}_2) \cdot \underline{l} = - \frac{d}{dt} \int_S \underline{\mu H} \cdot d\underline{a}$$

As the width of the loop $l \rightarrow 0$, the flux (RHS)

goes to 0

$$\Rightarrow \boxed{\underline{E}_1^\parallel = \underline{E}_2^\parallel}$$

$\Rightarrow$ Tangential component of electric field is continuous across interface

* If you remember nothing else from this lecture remember this BC as it is crucial in RS applications and will appear over and over

- Ampere's Law: Following the same logic as for Faraday's Law

$$\boxed{\underline{H}_1^\parallel = \underline{H}_2^\parallel} \quad (\text{Zero free current at surface})$$

$\Rightarrow$ Tangential component of H field is conserved

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• EM BCs (cont.)

- Summary: for RS application (no free charge) and lossless materials

4 BCs

$$1. \epsilon_1' E_1^\perp = \epsilon_2' E_2^\perp$$

→ $\epsilon E \equiv$ electric displacement $\Rightarrow$ displacement normal to interface is the same on both sides of interface

$$2. H_1^\perp = H_2^\perp$$

→ normal component of H field is same on both sides of the interface (assuming $\mu_1 = \mu_2$, common in RS as $\mu = \mu_0$ in nearly all cases)

$$* 3. E_1^\parallel = E_2^\parallel$$

→ tangential component of E field is conserved across interface

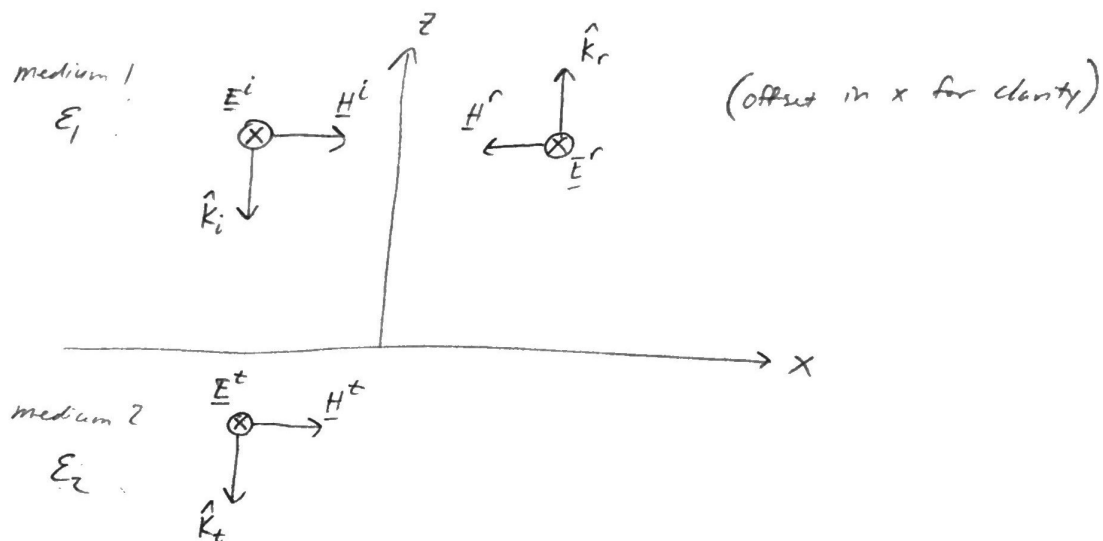
$$4. H_1^\parallel = H_2^\parallel$$

→ tangential component of H field is conserved across interface

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EM wave reflection and transmission at normal incidence

- Let's consider a simple and very useful example of normal incidence to a planar interface
- We have a uniform plane wave initially propagating downward $\hat{k}_i = -\hat{z}$
- Interface between 2 media at $z=0$



- E field is polarized in y-direction $\Rightarrow$ H is in x-direction define fields

$$\text{medium 1} \left\{ \begin{array}{l} \text{incident} \left\{ \begin{array}{l} \underline{E}^i = \hat{y} E_0^i e^{\gamma_1 z} \\ \underline{H}^i = \hat{x} \frac{\underline{E}_0^i}{\eta_1} e^{\gamma_1 z} \end{array} \right. \\ \text{reflected} \left\{ \begin{array}{l} \underline{E}^r = \hat{y} E_0^r e^{-\gamma_1 z} \\ \underline{H}^r = -\hat{x} \frac{\underline{E}_0^r}{\eta_1} e^{-\gamma_1 z} \end{array} \right. \end{array} \right. \quad \begin{array}{l} \gamma_1 = \alpha_1 + j\beta_1 \\ \eta_1 = \frac{\eta_0}{\sqrt{\epsilon_1}} \end{array}$$

$$\text{medium 2} \left\{ \begin{array}{l} \text{transmitted} \left\{ \begin{array}{l} \underline{E}^t = \hat{y} E_0^t e^{\gamma_2 z} \\ \underline{H}^t = \hat{x} \frac{\underline{E}_0^t}{\eta_2} e^{\gamma_2 z} \end{array} \right. \end{array} \right. \quad \begin{array}{l} \gamma_2 = \alpha_2 + j\beta_2 \\ \eta_2 = \frac{\eta_0}{\sqrt{\epsilon_2}} \end{array}$$

• Normal Incidence (cont.)

- Total fields in each medium are

$$\underline{E}_1 = \underline{E}^i + \underline{E}^r$$

$$\underline{H}_1 = \underline{H}^i + \underline{H}^r$$

$$\underline{E}_2 = \underline{E}^t$$

$$\underline{H}_2 = \underline{H}^t$$

- At $z=0$, must apply BCs from Maxwell's Eqs

- At normal incidence, fields are entirely tangential to interface (again, because EM waves always transverse)

$$\Rightarrow E_0^i + E_0^r = E_0^t \quad \rightarrow \text{tangential comp. of } E \text{ field conserved}$$

$$\text{and} \quad \frac{E_0^i}{\eta_1} - \frac{E_0^r}{\eta_1} = \frac{E_0^t}{\eta_2} \quad \rightarrow \text{tangential comp. of } H \text{ field conserved}$$

- In general in BS, know E_0^i , so solve for

$$\rho = \frac{E_0^r}{E_0^i} \quad \text{and} \quad \tau = \frac{E_0^t}{E_0^i}$$

- From above we see

$$1 + \rho = \tau \quad \text{and} \quad 1 - \rho = \frac{\eta_1}{\eta_2} \tau$$

- Solving,

$$\left. \begin{aligned} \rho &= \frac{\eta_2 - \eta_1}{\eta_2 + \eta_1} = \frac{\sqrt{\epsilon_1} - \sqrt{\epsilon_2}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}} \\ \tau &= \frac{2\eta_2}{\eta_2 + \eta_1} = \frac{2\sqrt{\epsilon_1}}{\sqrt{\epsilon_1} + \sqrt{\epsilon_2}} \end{aligned} \right\} \begin{array}{l} \text{Fresnel reflection } (\rho) \\ \text{and transmission } (\tau) \\ \text{coefficients for} \\ \text{normal incidence} \end{array}$$

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Review the (cont.)

- You may notice the similarity of forms between ρ and τ and reflection and transmission in strings

- Connecting to the string model given at the beginning of lecture, consider lossless media

($\epsilon_1 = \epsilon_1'$ and $\epsilon_2 = \epsilon_2'$) and recall that

$$v_{p1} = \frac{c}{\sqrt{\epsilon_1'}} \quad \text{and} \quad v_{p2} = \frac{c}{\sqrt{\epsilon_2'}}$$

$$\Rightarrow \left. \begin{aligned} \rho &= \frac{v_{p2} - v_{p1}}{v_{p2} + v_{p1}} = \frac{K_1 - K_2}{K_1 + K_2} \\ \tau &= \frac{2v_{p2}}{v_{p2} + v_{p1}} = \frac{2K_1}{K_1 + K_2} \end{aligned} \right\} \begin{array}{l} \text{identical to} \\ \text{reflection and} \\ \text{transmission coeffs} \\ \text{on string!} \end{array}$$

* - Key take-away point

reflected ^{EM} wave for general media

in phase w/ incident wave if

$$v_{p1} < v_{p2} \Rightarrow \operatorname{Re}(\sqrt{\epsilon_1'}) > \operatorname{Re}(\sqrt{\epsilon_2'})$$

out of phase (π offset) if

$$v_{p1} > v_{p2} \Rightarrow \operatorname{Re}(\sqrt{\epsilon_1'}) < \operatorname{Re}(\sqrt{\epsilon_2'})$$

→ virtually always the case when medium 1 is atmosphere

• Review the (cont.)

- We may also want to think about this in terms of the partitioning of energy between reflected and transmitted waves

$$\begin{aligned}\underline{S}_1 &= \frac{1}{2} \operatorname{Re} [\underline{E}_1 \times \underline{H}_1^*] \\ &= \frac{1}{2} \operatorname{Re} \left[\hat{y} E_0^i (e^{\gamma_1 z} + \rho e^{-\gamma_1 z}) \times \hat{x} \frac{E_0^{i*}}{\eta_1} (e^{-\gamma_1 z} - \rho^* e^{\gamma_1 z}) \right]\end{aligned}$$

at $z=0$ gives

$$\underline{S}_1 = -\hat{z} \frac{|E_0^i|^2}{2|\eta_1|} \underbrace{(1 - |\rho|^2)}_{\text{note this term!}} \cos \theta_{\eta_1}$$

$$= \underline{S}^i + \underline{S}^r$$

Similarly,

$$\begin{aligned}\underline{S}^t = \underline{S}_2 &= \frac{1}{2} \operatorname{Re} [\underline{E}_2 \times \underline{H}_2^*] \\ &= -\hat{z} |\tau|^2 \frac{|E_0^i|^2}{2|\eta_2|} \cos \theta_{\eta_2}\end{aligned}$$

From these expressions, can show

$$|\eta_1|^2 = \frac{|\eta_2| \cos \theta_{\eta_1}}{|\eta_1| \cos \theta_{\eta_2}} (1 - |\rho|^2)$$

and

$$\underline{S}_1 = \underline{S}_2 \quad (\text{sanity check: energy is conserved!})$$

Finally, common to define

$$R = \frac{S_0^r}{S_0^i} = |\rho|^2 \quad \text{and} \quad T = \frac{S_0^t}{S_0^i} = |\tau|^2 \frac{|\eta_1| \cos \theta_{\eta_2}}{|\eta_2| \cos \theta_{\eta_1}}$$

where it can be shown $R + T = 1$